



Automatic Light-Activated LED System Using LDR

A mini-project in **Analog Electronics (BTEC13403)** (**Sem-4, 2nd Year**) — an intelligent circuit that automatically switches an LED ON in darkness and OFF in light, using an Op-Amp comparator and an LDR sensor.

Guided by:-

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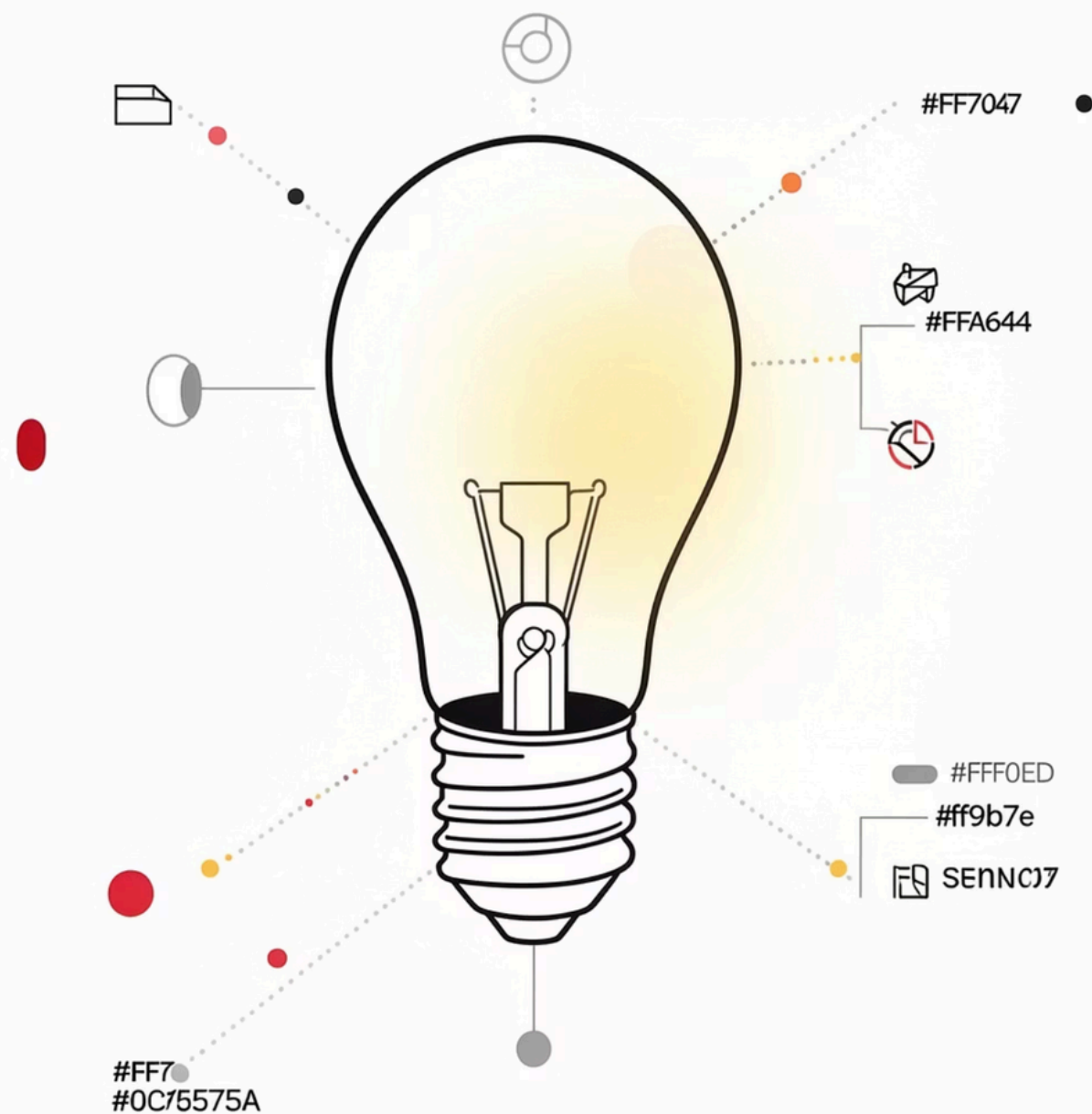
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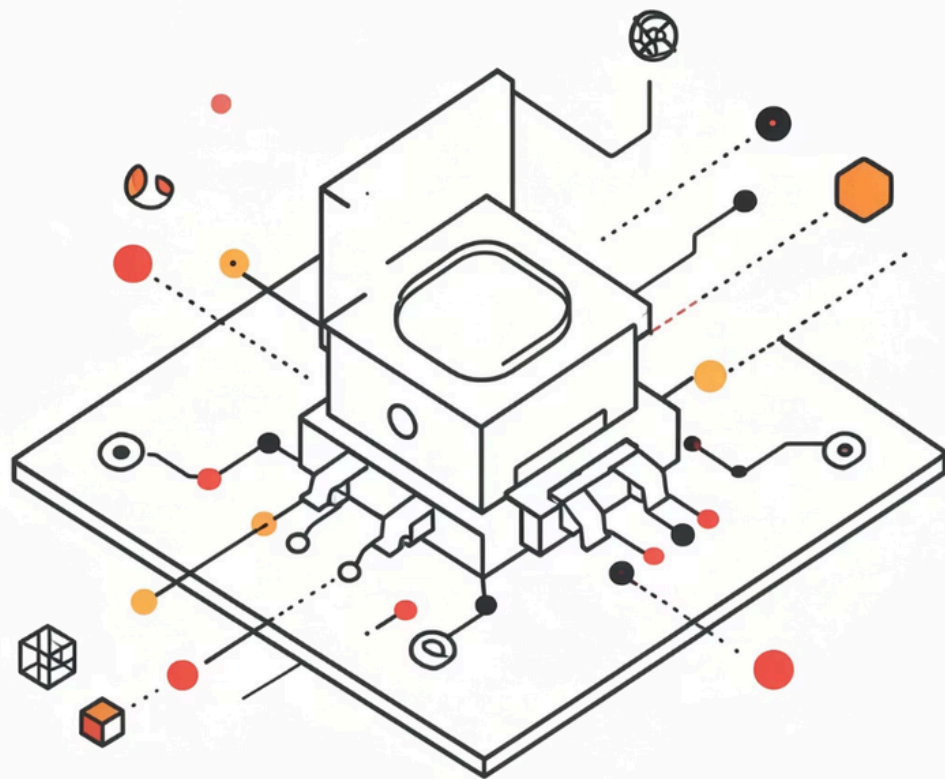
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About the Project!!



This project implements an **automatic ambient light detection circuit** using analog components. The core idea is simple but powerful: when the surrounding environment gets dark, the LED turns ON automatically — and when light is present, it turns OFF.

No microcontroller, no code — pure analog electronics using an Op-Amp 741 as a voltage comparator.

Components Used:-

Key parts of the automatic ambient light detection circuit are as follows:-

Op-Amp IC 741

Comparator between LDR & voltages

LDR Sensor

Detects ambient light level (RT in dark)

NPN Transistor

Acts as electronic switch for LED

Potentiometer – 10k Ω

Sets threshold/reference voltage

Resistor – 220 Ω

Current limiter for the LED protection

Resistor – 2.2k Ω

Base resistor for NPN transistor

Resistor – 22k Ω

Pull-up/voltage divider in LDR network

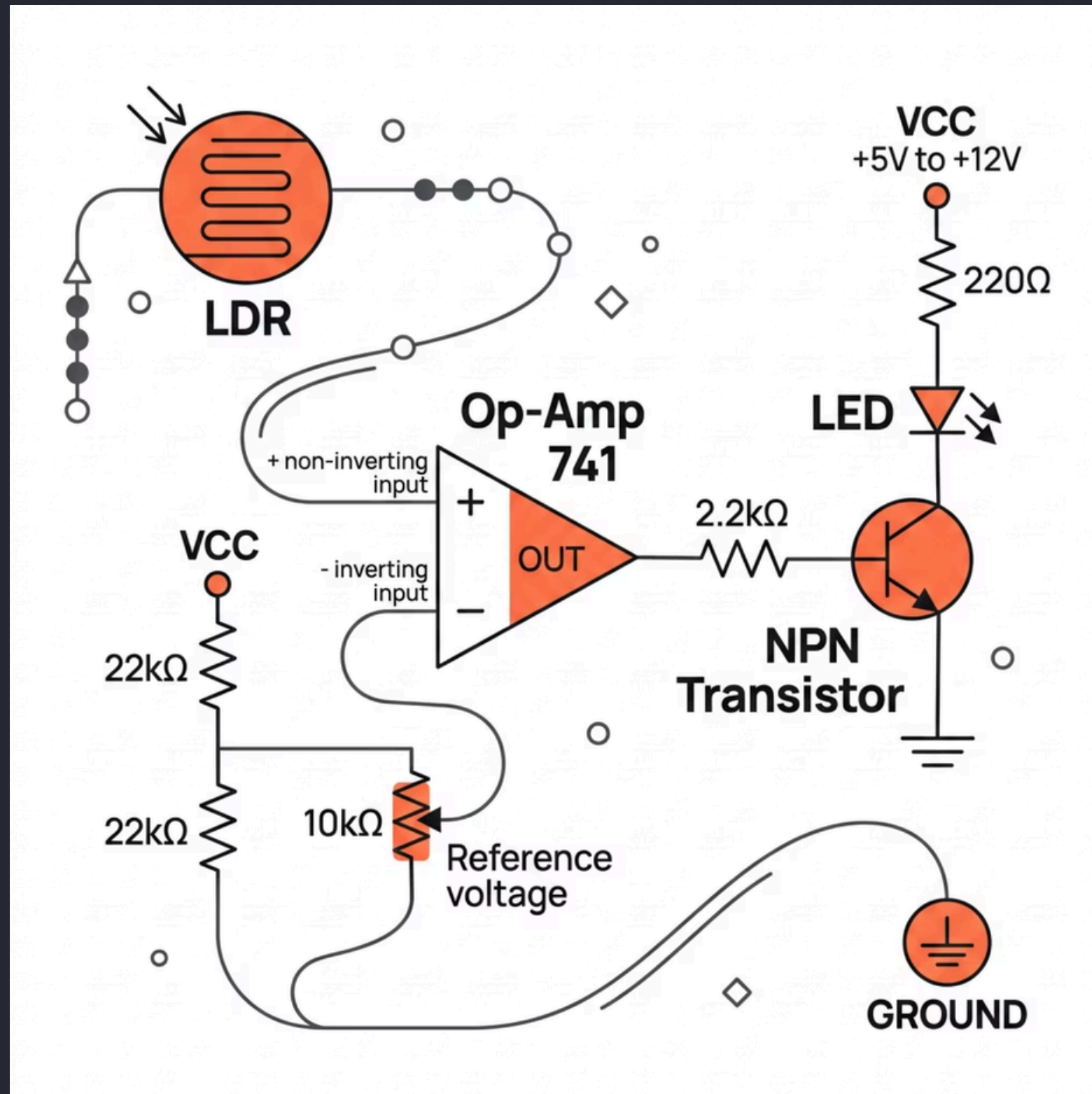
LED – RED

Output indicator — glows in darkness

Trainer Board & Wires

Prototyping platform & connections

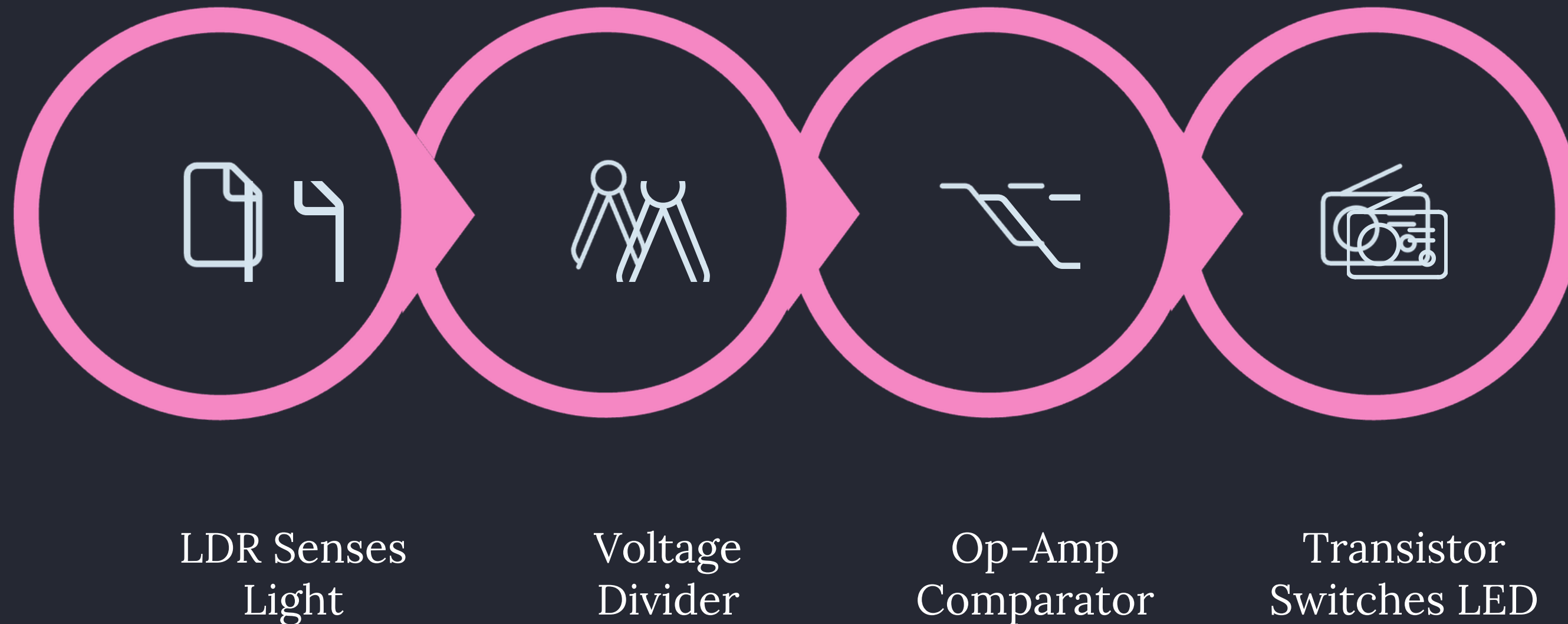
Circuit Diagram:-



How the Circuit is Wired?

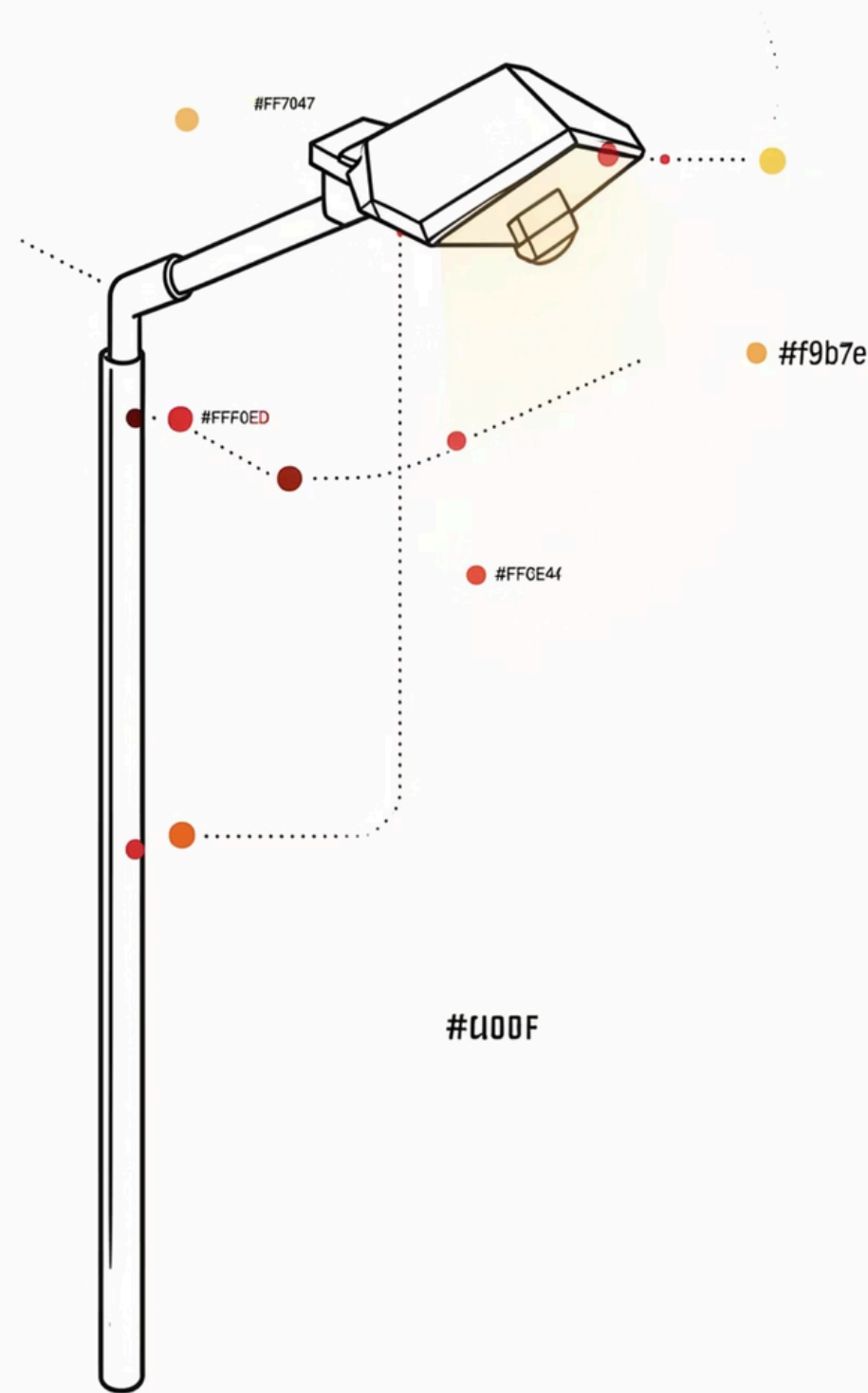
- **LDR + 22kΩ:-** form a voltage divider at the *non-inverting input (+)*.
- **10kΩ Potentiometer:-** sets the reference voltage at the *inverting input (-)*.
- **Op-Amp 741:-** compares both inputs and outputs, HIGH or LOW.
- **NPN Transistor:-** amplifies the output to drive the LED.
- **220Ω resistor:-** limits current through the LED.

Working Principle:-



The Op-Amp 741 operates in **open-loop comparator mode**. When LDR resistance rises in darkness, the voltage at the non-inverting input exceeds the reference, driving the output HIGH and turning the LED ON via the transistor switch.

Real-World Applications:-



Street Lighting

Automatically activate street lamps at dusk without manual intervention.

Home Automation

Corridor and outdoor lights that respond to ambient darkness.

Security Systems

Trigger indicator LEDs or alarms in low-light conditions.

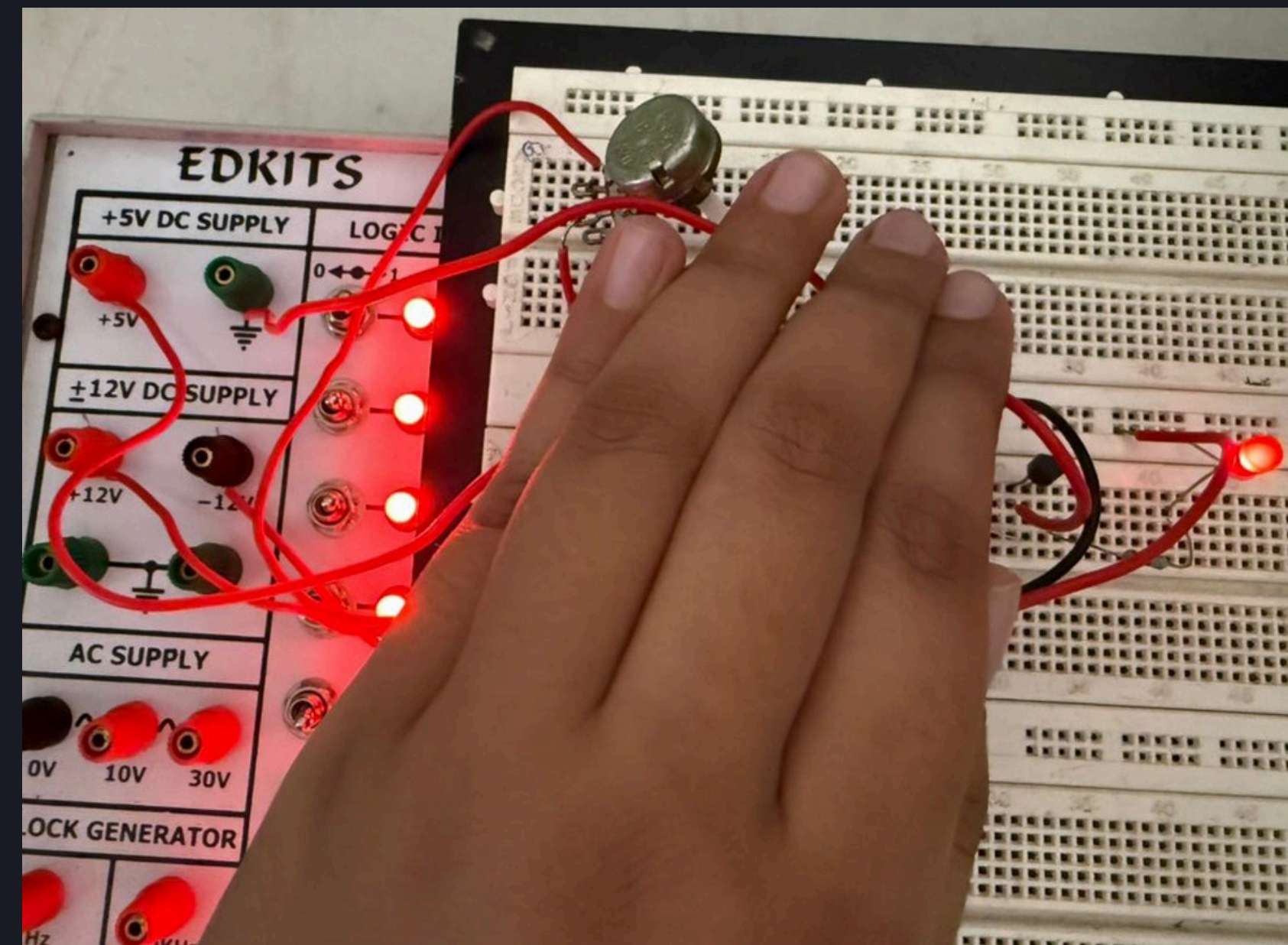
Greenhouses

Supplemental grow lights that activate when sunlight is insufficient.

Hardware Implementation:-

Step-by-Step Assembly:-

1. Place Op-Amp 741 on the trainer board, connect V+ and V- supply rails.
2. Build the LDR voltage divider (LDR + $22k\Omega$) to the non-inverting pin.
3. Connect the $10k\Omega$ potentiometer to set the reference voltage at the inverting pin.
4. Wire the Op-Amp output to the NPN transistor base via $2.2k\Omega$ resistor.
5. Connect LED + 220Ω resistor from collector to VCC.
6. Power the circuit (5V–12V) and adjust potentiometer for desired sensitivity.



Advantages:-

Zero Programming Required

Purely analog — no microcontroller, Arduino, or code needed. Ideal for learning fundamental electronics.

Low Cost & Readily Available

All components (Op-Amp 741, LDR, transistor) are inexpensive and commonly found in any electronics lab.

Adjustable Sensitivity

The 10k Ω potentiometer allows fine-tuning of the light threshold for different environments.

Energy Efficient

LED only draws power when darkness is detected, reducing unnecessary energy consumption.

Limitations & Future Scope:-

Limitations:-

- Sensitive to ambient light fluctuations — may flicker near threshold
- No hysteresis — lacks noise immunity without Schmitt trigger modification
- Limited to single LED output — not scalable without additional driver stages
- Performance depends on power supply stability

Future Enhancements:-

- Add **hysteresis** using positive feedback for stable switching
- Replace LED with a **relay module** to control high-power lamps
- Integrate with **solar panels** for autonomous outdoor lighting
- Use multiple LDRs for **directional light sensing**
- Upgrade to a **comparator IC (LM311)** for better precision

Conclusion:-

This mini-project demonstrates the practical application of analog electronics concepts — voltage comparison, sensor interfacing, and transistor switching — in a single, functional circuit.

Core Concept

Op-Amp as comparator + LDR as sensor

Key Outcome

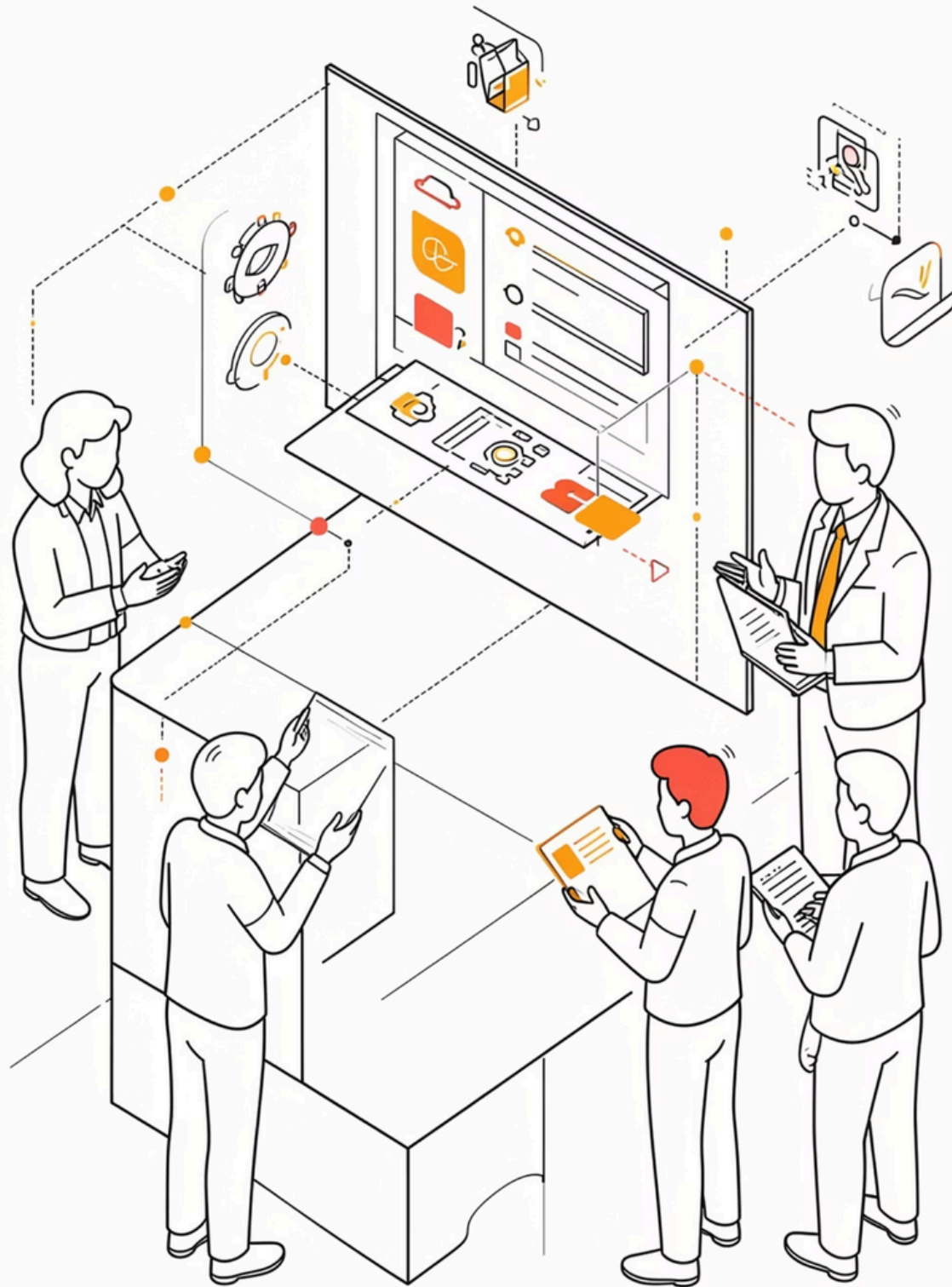
LED auto-activates in darkness

Learning Value

Hands-on analog circuit design

Scalability

Foundation for smart lighting systems



Thank You!!